IIITB COMET FOUNDATION

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QUESTION

PH

The following Boolean expression

$Y = A \bullet \overline{B} \bullet \overline{C} \bullet \overline{D} + \overline{A} \bullet B \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet C \bullet D + \overline{A} \bullet B \bullet C \bullet D + A \bullet \overline{B} \bullet \overline{C} \bullet D$

can be simplified to

(A) $\overline{A} \bullet \overline{B} \bullet C + A \bullet \overline{D}$

(B) $\overline{A} \bullet B \bullet \overline{C} + A \bullet \overline{D}$

(C) $A \bullet \overline{B} \bullet \overline{C} + \overline{A} \bullet D$

(D) $A \bullet \overline{B} \bullet C + \overline{A} \bullet D$

QUESTION ANALYSIS

Solution

can be simplified to

(A) $\overline{A} \cdot \overline{B} \cdot C + A \cdot \overline{D}$ **(B)** $\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{D}$

(C) $A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot D$ **(D)** $A \cdot \overline{B} \cdot C + \overline{A} \cdot D$

Solution

Given,

$$Y = A\overline{BCD} + \overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}BCD + \overline{A}BCD + A\overline{BC}D$$

Step 1: Group the terms

Group the first and last terms:

$$A\overline{BCD} + A\overline{BC}D = A\overline{BC}(\overline{D} + D)$$

Since $\overline{D} + D = 1$,

$$A\overline{BC}(\overline{D} + D) = A\overline{BC}.$$

Now group the remaining four terms:

$$\overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}BCD + \overline{A}BCD$$

$$= \overline{A}D\left(B\overline{C} + \overline{BC} + \overline{B}C + BC\right).$$

Step 2: Simplify inside the brackets

$$
\begin{aligned}
& B\overline{C} + \overline{B\overline{C}} + \overline{B}C + BC \\
&= \overline{C}(B + \overline{B}) + C(B + \overline{B}) \\
&= \overline{C}(1) + C(1) \\
&= \overline{C} + C \\
&= 1.
\end{aligned}
$$

Therefore,

$$\overline{A}D(1) = \overline{A}D.$$

Step 3: Final expression

Hence,

$$\boxed{Y = A\overline{B\overline{C}} + \overline{A}D}$$

This matches **Option (C).**

$$\boxed{\text{Correct Answer: (C)}}$$

Hardware Components Required

S.No.	Component
1	ESP32-WROOM Development Board
2	Personal Computer (Ubuntu OS)
3	USB Cable
4	Breadboard
5	7474 Dual D Flip-Flop IC -2(number)
6	7447 BCD-to-7-Segment Decoder IC
7	Common Anode 7-Segment Display
8	220 Ω Resistors (7 Nos.)
9	Jumper Wires

Table 1: Hardware and Software Components Required for the Experiment

Hardware Connections

Table 1: Connections Between ESP32-WROOM and 7474 IC

S.No.	ESP32-WROOM Pin	7474 IC Pin/Signal
1	GPIO32	D Input (Flip-Flop A)
2	GPIO33	D Input (Flip-Flop B)
3	GPIO21	Clock (CLK)
4	VIN (5V)	VCC (Pin 14)
5	GND	GND (Pin 7)
6	GPIO25	PRE (Preset)
7	GPIO26	CLR (Clear)

Table 2: Connections Between ESP32-WROOM and 7474 IC

Table 2: Connections Between 7474 IC and 7447 IC

S.No.	7474 Output	7447 Input
1	QA	A
2	QB	B
3	GND	C (Logic 0)
4	GND	D (Logic 0)

Table 3: Connections Between 7474 IC and 7447 IC

Table 3: Connections Between 7447 IC and Common Anode 7-Segment Display

S.No.	7447 Output	7-Segment Display
1	a	Segment a
2	b	Segment b
3	c	Segment c
4	d	Segment d
5	e	Segment e
6	f	Segment f
7	g	Segment g
8	Common Anode	+5 V Supply
9	Segment Pins	Through 220 Ω Resistors

Table 4: Connections Between 7447 IC and Common Anode 7-Segment Display

S.No.	Component	Connection
1	ESP32 USB Port	Ubuntu PC USB Port
2	USB Cable	Power and Program Upload
3	Ubuntu PC	PlatformIO and ESPTool

Table 5: Programming Connections

Table 3: Programming Connections

Execution Procedure

1. Create a new PlatformIO project for the ESP32-WROOM board.

   ```
   /root/venv/bin/pio project init --board esp32dev
   ```

2. Place the source code (`main.cpp`) inside the `src` directory of the project.

3. Compile the source code and generate the binary files using:

   ```
   /root/venv/bin/pio run
   ```

4. After successful compilation, the following binary files are generated in the build directory:

   `.pio/build/esp32dev/`

   - `bootloader.bin`
   - `partitions.bin`
   - `firmware.bin`

5. Copy the generated binary files to the desired directory on the Ubuntu PC.

6. Connect the ESP32-WROOM board to the Ubuntu PC using a USB cable.

7. Verify that the ESP32 board is detected by the operating system.

   ```
   ls -l /dev/ttyUSB*
   ```

8. Verify the ESP32 chip.

   ```
   esptool.py --chip esp32 --port /dev/ttyUSB0 chip_id
   ```

9. Flash the generated binary files into the ESP32.

```
esptool.py --chip esp32 \
  --port /dev/ttyUSB0 \
  --baud 460800 \
  write_flash \
  0x1000 bootloader.bin \
  0x8000 partitions.bin \
  0x10000 firmware.bin
```

10. Wait until the flashing process completes successfully.
11. Press the **EN (Reset)** button on the ESP32 board if required.
12. Observe the LED connected to the output GPIO pin to verify the implemented Boolean expression.

Results and Observations

The Boolean function was successfully implemented using the ESP32-WROOM development board, 7474 D Flip-Flop IC, 7447 BCD-to-7-Segment Decoder IC, and a common anode 7-segment display. The ESP32 generated the required input combinations and clock signal for the 7474 IC. The output stored in the flip-flop was decoded by the 7447 IC and displayed on the 7-segment display.

Table 6 presents the truth table of the implemented Boolean function. For every input combination, the observed output matched the expected output.

Truth Table

A	B	C	D	Y	7-Segment Display
0	0	0	0	0	0
0	0	0	1	1	1
0	0	1	0	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

Table 6: Truth Table of the Implemented Boolean Function

Figure 1 shows the experimental setup displaying the output of the implemented Boolean function. The 7-segment display indicates the digit '0' when the Boolean output is logic LOW ($Y = 0$) and the digit '1' when the Boolean output is logic HIGH ($Y = 1$).

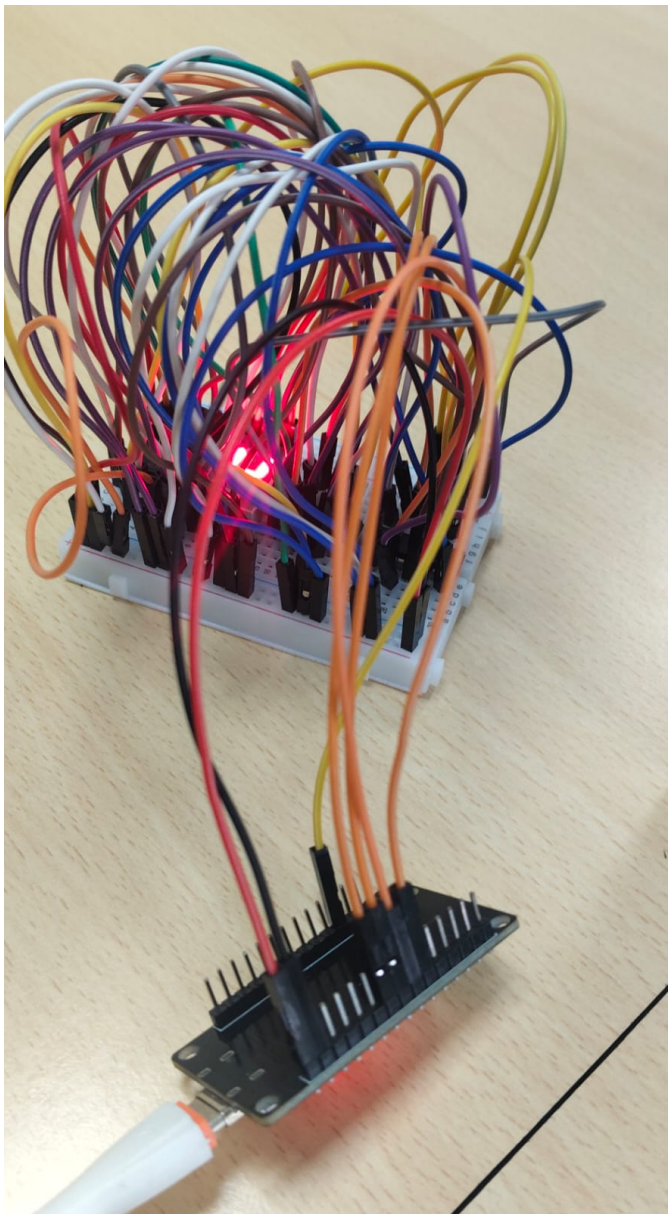


Figure 1: Experimental output displayed on the common anode 7-segment display.

Result

The Boolean function was successfully implemented and verified using the ESP32-WROOM, 7474 D Flip-Flop IC, 7447 BCD-to-7-Segment Decoder IC, and common anode 7-segment display. The experimental results were in agreement with the theoretical truth table, confirming the correct operation of the designed circuit.